INVERSE SOURCE PROBLEM FOR THE STOCHASTIC FRACTIONAL HELMHOLTZ EQUATION IN INHOMOGENEOUS MEDIA

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Abstract. This paper is concerned with an inverse source problem for the stochastic fractional Helmholtz equation in a three-dimensional inhomogeneous medium. The source is modeled as real Gaussian spatial white noise with an unknown nonnegative strength function. For admissible frequencies, we establish well-posedness of the radiating solution in the distributional sense and show that it has a smooth representation outside the source region. For the inverse problem, by using two-point correlations of the field measured in an exterior open set at a single frequency, we prove a logarithmic stability estimate for recovering the source strength under an a priori Sobolev bound. As a uniform Hölder stability estimate is unavailable for the general admissible class, by assuming the difference between two source strengths has a fixed sign, we improve the stability to Hölder continuity with an optimal Hölder exponent. The analysis combines scattering theory and quantitative continuation for the fractional Laplacian using the Caffarelli-Silvestre Extension.

1. Introduction

The fractional Helmholtz operator arises in fixed-energy models of fractional quantum mechanics [11] and in descriptions of spatial dispersion [23]. Its nonlocality couples the field across separated points. In an inhomogeneous medium, the refractive coefficient and potential also affect wave propagation. When the source fluctuates randomly, correlations of the radiated field provide information about its statistical properties.

We study the inverse source problem for the stochastic fractional Helmholtz equation

$$((-\Delta)^{\sigma} - k^{2\sigma} r(x) + q(x))u(x,k) = \sqrt{\mu(x)}\,\dot{W}(x) \quad \text{in } \mathbb{R}^3, \tag{1.1}$$

where $0 < \sigma < 1$, $k > 0$, r is a refractive coefficient, and q is a potential. The fractional Laplacian is defined by the Fourier multiplier

$$\widehat{(-\Delta)^{\sigma} v}(\xi) = |\xi|^{2\sigma} \widehat{v}(\xi).$$

The functions $1 - r$ and q are compactly supported, while $\dot{W}$ denotes real Gaussian spatial white noise. The nonnegative function μ describes the strength of the random source. We assume that the source and the inhomogeneities are contained in K with $K \Subset B$, where B is a bounded region. We seek the outgoing field satisfying the Sommerfeld radiation condition:

$$\lim_{\rho \to \infty} \rho(\partial_\rho u - iku) = 0, \qquad \rho = |x|, \tag{1.2}$$

uniformly in the direction $\hat{x} = x/|x|$.

Our objective is to determine μ from measurements made in an open set O outside B. More precisely, the data are the two-point correlations

$$F(x, z; k) = \mathbb{E}[u(x,k)u(z,k)], \qquad x, z \in O, \tag{1.3}$$

at a fixed frequency k. The geometric assumptions are specified at the beginning of Section 2 and apply throughout the paper. Because the source is spatial white noise, its individual realizations

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are distributions rather than ordinary functions. Establishing the meaning and regularity of the radiating solution is therefore a necessary part of the inverse analysis.

The known coefficients $r, q \in C^\infty(\mathbb{R}^3;\mathbb{R})$ are fixed and satisfy

$$r \geq r_0 > 0, \qquad \operatorname{supp}(r-1) \cup \operatorname{supp} q \subset K.$$

Our main result is formulated as follows. Let u_1, u_2 be the solutions corresponding to two sources $\sqrt{\mu_1}\dot{W}$ and $\sqrt{\mu_2}\dot{W}$, respectively. The data discrepancy is defined by

$$\delta = \|F_1 - F_2\|_{L^2(O\times O)}, \tag{1.4}$$

where $F_i = \mathbb{E}[u_i(x)u_i(z)], i = 1, 2$. For $m > 3/2$ and $E > 0$, introduce the a priori class for the strength functions

$$\mathcal{A} = \{\mu \in H^m(\mathbb{R}^3) : \mu \geq 0,\ \operatorname{supp}\mu \subset K,\ \|\mu\|_{H^m} \leq E\}. \tag{1.5}$$

According to Theorem 2.3, the direct problem for the fractional Helmholtz equation with variable coefficients is well posed for all $k > 0$ except for $k \in \Sigma \subset (0, +\infty)$, where Σ consists of discrete poles. Thus, for the study of the inverse problem we assume $k \notin \Sigma$.

The following stability estimate is the main result of this paper, which implies the uniqueness of the inverse problem.

Theorem 1.1. *Assume $k \notin \Sigma$. Let $0 < \sigma < 1$ and $m > 3/2$. For $\mu_1, \mu_2 \in \mathcal{A}$, the following stability estimate holds*

$$\|\mu_1 - \mu_2\|_{L^\infty} \leq CE\left(\log\left(3 + \frac{E}{\delta}\right)\right)^{-\theta}, \tag{1.6}$$

where

$$\theta = \theta_0 \frac{m - 3/2}{\max\{m - 1 + \sigma, 1\}} > 0. \tag{1.7}$$

Here $C > 0$ is a constant, and θ_0 may be any fixed exponent with $0 < \theta_0 < \sigma/\kappa$, where $\kappa > 0$ is the constant specified by Lemma 3.1.

The logarithmic stability obtained above in general cannot be improved to a Hölder one. To obtain a Hölder stability, we assume that the difference of the strength functions has a fixed sign. As a consequence, we obtain the following Hölder stability.

Theorem 1.2. *Assume $0 < \sigma < 1$, $m > 3/2$, and $k \notin \Sigma$. Let $\mu_1, \mu_2 \in \mathcal{A}$ satisfy either $\mu_1 \geq \mu_2$ or $\mu_2 \geq \mu_1$. One has the stability estimates*

$$\|\mu_1 - \mu_2\|_{L^1} \leq C\delta \quad \text{and} \quad \|\mu_1 - \mu_2\|_{L^\infty} \leq CE^{1-\beta}\delta^\beta, \tag{1.8}$$

where C is a generic constant and

$$\beta = \frac{m - 3/2}{m + 3/2}.$$

Moreover, the exponent β is optimal.

Unique continuation has played a central role in inverse problems for fractional operators. The fractional Calderón problem and its stability have been studied in [10, 21, 20, 6]. For deterministic fractional Helmholtz equations, simultaneous recovery of a source and a scattering potential was studied in [3]. High-frequency inverse scattering in variable refractive backgrounds was investigated in [4], and deterministic potential recovery from far-field scattering amplitudes was obtained in [7].

The Caffarelli–Silvestre extension [2] realizes $(-\Delta)^\sigma$ as the weighted Dirichlet-to-Neumann operator of a local degenerate elliptic equation in one higher dimension. This local–nonlocal correspondence extends to fractional powers of second-order operators [22]. In fractional inverse problems, the extension has supported Carleman-based unique continuation [19], partial-data recovery through Runge approximation [10, 9], and quantitative propagation of smallness for stability [21, 20]. It also provides a route, under the variable-coefficient assumptions of [5], from partial nonlocal Calderón

data to the full local Dirichlet-to-Neumann map. Related extension-based reductions for elliptic inverse source and parabolic Calderón problems appear in [15, 14]. In the present proof, we use the extension to propagate smallness of an auxiliary field from the exterior observation region toward the source region.

For classical stochastic Helmholtz equations, inverse random-source scattering was studied in [1]; stability for white-noise sources, including in inhomogeneous media, was established in [12]; and fixed-frequency conditional stability was obtained in [17]. In the fractional setting, [13] recovers principal symbols of the covariance and relation operators of a microlocally isotropic generalized Gaussian source from a single realization of high-frequency far-field data. Its model includes a known deterministic potential supported away from the source. For a homogeneous fractional white-noise model, [18] proves single-frequency uniqueness for the variance and obtains increasing stability from multifrequency exterior correlations.

In this work, the unknown is the strength μ of a real spatial white-noise source. The known inhomogeneities may overlap its support, and the data consist of ensemble correlations on $O \times O$ at one admissible frequency. Our aim is a quantitative stability estimate for recovering the strength function. To this end, we first establish a direct theory using the outgoing free resolvent and an integral equation formulation. For admissible frequencies, the solution exists in the distributional sense and has a smooth Green function representation in the exterior observation region. When $\sigma > 3/4$, it also has a continuous mild realization, so the stochastic integral formulation can be interpreted pointwise throughout the space. For strengths in a bounded $H^m(\mathbb{R}^3)$ class, $m > 3/2$, we bound the L^∞ difference of the strength functions logarithmically by the discrepancy of the exterior correlations at a fixed frequency. This implies uniqueness. We also show that a uniform Hölder stability estimate does not hold for the full admissible class. If the difference of two strengths has a fixed sign, we obtain a Lipschitz estimate in L^1 and a Hölder estimate in L^∞ with an optimal exponent.

The stability analysis starts from an identity expressing the field correlation as an integral of μ against a product of Green functions. An approximation property for fields generated from the observation region transfers the data discrepancy to the source region. Quantitative continuation for the fractional Laplacian [21] then gives the $H^{1-\sigma}$ control needed for logarithmic L^∞ stability. Under the fixed-sign assumption, a positive test of the correlation identity yields the L^1 estimate, and interpolation gives the optimal Hölder exponent.

The paper is organized as follows. Section 2 establishes the direct problem and the continuous mild realization. Section 3 proves the stability estimates and examines their sharpness. In the appendix we collect the continuation and approximation results used in the inverse analysis.

2. Direct problem

Throughout the paper, including the statements in the Introduction, B is a bounded open set, $K \Subset B$ is compact with nonempty interior, and O is a nonempty bounded open subset of $\mathbb{R}^3 \setminus \overline{B}$. These assumptions imply $\operatorname{dist}(K, \overline{O}) > 0$. For the inverse results, the source strengths belong to $\mathcal{A}$ in (1.5). We use the Fourier transform $\widehat{f}(\xi) = \int_{\mathbb{R}^3} e^{-ix\cdot\xi} f(x)\, dx$, whose inverse carries the factor $(2\pi)^{-3}$. For the inverse estimates, $k \notin \Sigma$ is fixed. Their constants may depend on $\sigma, m, k, r, q, K, B, O$ and the chosen θ_0. The exponent θ_0 is fixed with $0 < \theta_0 < \sigma/\kappa$, where κ is provided by Lemma 3.1, with the observation ball chosen once and for all inside O. Every logarithmic modulus is interpreted as zero when its discrepancy argument is zero.

We make precise the distributional interpretation and the outgoing solution class. A candidate solution is a tempered distribution u whose Fourier transform is locally integrable near $\xi = 0$. We require u to be smooth outside a ball and to satisfy $u(x) = O(|x|^{-1})$ and (1.2), uniformly in $x/|x|$. Near zero the product $|\xi|^{2\sigma}\widehat{u}$ is ordinary function multiplication; away from zero it is distributional multiplication by a smooth symbol. A smooth cutoff combines these definitions, which agree on

their overlap. All distributional solutions and uniqueness assertions below are understood in this class. For random fields, these conditions hold on one event of probability one.

2.1. **Well-posedness of the direct problem.** We first consider the free equation corresponding to $r = 1$ and $q = 0$. The outgoing free resolvent is denoted by

$$\mathcal{R}_0(k) = \left((-\Delta)^\sigma - k^{2\sigma} - i0\right)^{-1}.$$

Its convolution kernel $G_{\sigma,k}$ has the following properties (cf. [18, Lemma 2.1]).

Lemma 2.1. *Let $0 < \sigma < 1$ and $k > 0$. Then*

$$G_{\sigma,k}(x) = \frac{k^{2-2\sigma}}{4\pi\sigma}\frac{e^{ik|x|}}{|x|} + J_{\sigma,k}(|x|), \tag{2.1}$$

$$J_{\sigma,k}(d) = \frac{\sin(\pi\sigma)}{4\pi^2 d}\int_0^\infty \frac{t^\sigma e^{-d\sqrt{t}}}{t^{2\sigma} - 2k^{2\sigma}t^\sigma\cos(\pi\sigma) + k^{4\sigma}}\,dt. \tag{2.2}$$

The kernel is smooth away from zero and locally integrable, with

$$G_{\sigma,k}(x) = d_\sigma|x|^{2\sigma-3}(1 + o(1)) \quad (|x| \to 0), \qquad d_\sigma = \frac{\sin(\pi\sigma)\Gamma(2-2\sigma)}{2\pi^2} > 0. \tag{2.3}$$

Every derivative of $J_{\sigma,k}(|x|)$ of order j is $O(|x|^{-3-2\sigma-j})$ at infinity. For $\chi_1, \chi_2 \in C_c^\infty(\mathbb{R}^3)$ and $t \in \mathbb{R}$, the localized resolvent operator

$$\chi_1\mathcal{R}_0(k)\chi_2 : H^t(\mathbb{R}^3) \longrightarrow H^{t+2\sigma}(\mathbb{R}^3) \tag{2.4}$$

is bounded. Moreover, the localized operator $\chi_1\mathcal{R}_0(z)\chi_2$ extends holomorphically to $\Re z > 0$ in the complex plane using the principal branch of $z^{2\sigma}$.

Proof. The Green kernel decomposition follows from the fractional-power resolvent identity in [16, (5.28)]; the explicit formulas and their local and far-field estimates are given in [18, Lemma 2.1]. We give the operator argument for the last assertions. Extend the kernel decomposition to $\Re z > 0$ using the principal branch. The Fourier multiplier of its tail term is

$$\frac{\sin(\pi\sigma)}{\pi}\int_0^\infty \frac{t^\sigma\,dt}{(t + |\xi|^2)(t^{2\sigma} - 2t^\sigma z^{2\sigma}\cos(\pi\sigma) + z^{4\sigma})}.$$

On every compact subset of this half-plane the denominator's second factor has modulus at least $c(t^{2\sigma} + |z|^{4\sigma})$. Indeed, it factors as $(z^{2\sigma} - t^\sigma e^{i\pi\sigma})(z^{2\sigma} - t^\sigma e^{-i\pi\sigma})$, whereas $|\arg z^{2\sigma}| < \pi\sigma$. Splitting the integral at $t = |\xi|^2$ for $|\xi| \geq 1$, and using $t^{\sigma-1}$ as an integrable majorant near zero otherwise, we bound this multiplier by $C\langle\xi\rangle^{-2\sigma}$ locally uniformly in z. The scalar Cauchy formula, with this bound, also gives holomorphy in the operator norm from H^t to $H^{t+2\sigma}$. The classical Helmholtz resolvent after spatial localization is entire from H^t to H^{t+2}. Its compactly localized kernel is $e^{iz|x-y|}/(4\pi|x-y|)$ and the usual local elliptic estimate gives the stated two-derivative regularity gain. Multiplication by $z^{2-2\sigma}/\sigma$ preserves holomorphy in the right half-plane. The sum agrees with the spectral resolvent in the first quadrant and with its outgoing boundary value for real $k > 0$. This proves the localized mapping property and continuation. The proof is complete. □

Set

$$V = q + k^{2\sigma}(1 - r). \tag{2.5}$$

For a compactly supported source f, the corresponding Lippmann–Schwinger equation is

$$u = \mathcal{R}_0(k)(f - Vu).$$

The outgoing resolvent for the variable-coefficient equation is denoted by $\mathcal{R}(k) = ((-\Delta)^\sigma + q - k^{2\sigma}r)^{-1}$.

The Lippmann–Schwinger formulation and discrete exceptional frequencies for deterministic fractional Helmholtz scattering are also studied in [24, Theorems 4.1 and 4.2]. In the following lemma,

we prove the mapping and uniqueness statements needed here for compactly supported distributional sources.

Lemma 2.2. *There is a discrete set $\Sigma \subset (0,\infty)$ such that, for $k \notin \Sigma$, every compactly supported source has a unique outgoing solution in the class specified above. The resolvent satisfies*

$$\mathcal{R}(k) : H^t_{\mathrm{comp}}(\mathbb{R}^3) \longrightarrow H^{t+2\sigma}_{\mathrm{loc}}(\mathbb{R}^3), \qquad t \in \mathbb{R}. \tag{2.6}$$

The bounds are uniform when the source support and the localization sets are fixed. Its Green kernel $\mathcal{G}_k(x,y)$ is symmetric in x, y and smooth away from $x = y$.

Proof. Choose $\chi \in C^\infty_c(\mathbb{R}^3)$ equal to one near $\overline{B}$, and for $\Re z > 0$ put

$$T = I + \chi \mathcal{R}_0(z) V, \qquad V = q + z^{2\sigma}(1-r).$$

For every $s \in \mathbb{R}$, Lemma 2.1 shows that $T - I$ maps H^s boundedly into $H^{s+2\sigma}$, with output supported in $\operatorname{supp}\chi$. The compact Sobolev embedding therefore makes it a holomorphic compact family on H^s. In particular, T is Fredholm of index zero on each of these spaces.

First take z in the first quadrant and work on L^2. If $Tv = 0$, then $v = -\chi\mathcal{R}_0(z)Vv$, and repeated use of the 2σ-derivative gain shows that v is smooth and compactly supported. Put $w = -\mathcal{R}_0(z)Vv$. Since $\chi w = v$ and $\chi = 1$ on $\operatorname{supp} V$,

$$((-\Delta)^\sigma + q - z^{2\sigma} r)w = 0.$$

The free spectral resolvent maps L^2 to $H^{2\sigma}$ globally there, so $w \in H^{2\sigma}(\mathbb{R}^3)$. Multiplying the equation by $\overline{w}$, integrating over $\mathbb{R}^3$, and taking imaginary parts gives

$$-\Im(z^{2\sigma}) \int_{\mathbb{R}^3} r|w|^2 \, dx = 0.$$

Since $\Im(z^{2\sigma}) > 0$ and $r \geq r_0 > 0$, this implies $w = v = 0$. Thus T is invertible there. The analytic Fredholm theorem gives a meromorphic inverse on the right half-plane. We denote by Σ its positive real poles.

At any $k \notin \Sigma$, a vector in the kernel of T on an arbitrary H^s is compactly supported and also belongs to L^2 by the repeated regularity gain argument. Its vanishing follows from the L^2 inverse. Index zero then gives a bounded inverse on every H^s. These inverses agree on their common domains, since their difference lies in the kernel on the lower of the two Sobolev spaces. Consequently the same exceptional set works for every Sobolev order.

For $f \in H^t$ with compact support, define

$$v = T^{-1}\chi\mathcal{R}_0(k)f, \qquad u = \mathcal{R}(k)f = \mathcal{R}_0(k)(f - Vv),$$

using the inverse on $H^{t+2\sigma}$. The defining equation gives $\chi u = v$ and hence $Vu = Vv$. These identities and Lemma 2.1 prove (2.6). We also have $u = \mathcal{R}_0(k)(f - Vu)$. The effective source $f - Vu$ is a compactly supported distribution. Its Fourier transform is smooth and of polynomial growth, so

$$\widehat{u} = (|\xi|^{2\sigma} - k^{2\sigma} - i0)^{-1}(\widehat{f - Vu})$$

is locally integrable near zero and verifies the equation in the specified distributional sense. The kernel decomposition in Lemma 2.1, including source-variable derivatives of any fixed order, gives uniformly for y in a fixed compact set

$$\partial_y^\alpha G_{\sigma,k}(x-y) = O(|x|^{-1}), \qquad \partial_y^\alpha(\partial_{|x|} - ik)G_{\sigma,k}(x-y) = O(|x|^{-2}).$$

Pairing these estimates with the compactly supported finite-order distribution $f - Vu$ proves that u belongs to the outgoing class.

Conversely, let u be any outgoing solution with compactly supported source f. Then

$$h = u - \mathcal{R}_0(k)(f - Vu)$$

belongs to the same class and satisfies $((-\Delta)^\sigma - k^{2\sigma})h = 0$. Near zero this Fourier equation is an equation of locally integrable functions and yields $\widehat{h} = 0$ there. Away from zero, it gives $\operatorname{supp}\widehat{h} \subset \{|\xi| = k\}$. In a neighborhood of this sphere the quotient

$$\frac{|\xi|^2 - k^2}{|\xi|^{2\sigma} - k^{2\sigma}}$$

extends smoothly and is nonzero. Multiplication by this quotient therefore yields $(-\Delta - k^2)h = 0$ on all of $\mathbb{R}^3$. Thus h is a smooth entire Helmholtz solution satisfying the Sommerfeld condition, and classical Rellich uniqueness gives $h = 0$. Every outgoing solution hence satisfies the Lippmann–Schwinger equation and its restriction χu satisfies the invertible equation for T, proving uniqueness.

To prove reciprocity, take $f, g \in C_c^\infty(\mathbb{R}^3)$ and first let z lie in the first quadrant. The fields $\mathcal{R}(z)f$ and $\mathcal{R}(z)g$ belong to $H^{2\sigma}(\mathbb{R}^3)$. Bilinear integration by parts for the even Fourier symbol $|\xi|^{2\sigma}$ and for the multiplication operator $q - z^{2\sigma}r$ gives $\int f\mathcal{R}(z)g = \int g\mathcal{R}(z)f$. Both sides are meromorphic functions of z, by the localized construction, so continuation gives

$$\int_{\mathbb{R}^3} f\,\mathcal{R}(k)g\,dx = \int_{\mathbb{R}^3} g\,\mathcal{R}(k)f\,dx, \qquad f, g \in C_c^\infty(\mathbb{R}^3), \quad k \notin \Sigma. \tag{2.7}$$

This is a bilinear identity, and implies the symmetry of the distribution kernel.

Finally, let χ_1, χ_2 be compactly supported smooth cutoffs with disjoint supports, take $f = \chi_2 h$, and set $u = \mathcal{R}(k)f$. Choose a cutoff ψ supported away from $\operatorname{supp}\chi_2$ and equal to one near $\operatorname{supp}\chi_1$. The Lippmann–Schwinger equation gives

$$u = \mathcal{R}_0(k)f - \mathcal{R}_0(k)(V\psi u) - \mathcal{R}_0(k)(V(1-\psi)u).$$

Near $\operatorname{supp}\chi_1$, the first and third terms have smooth kernels and their Sobolev norms of every order are controlled by a fixed negative Sobolev norm of h, using (2.6) for Vu. The middle term gains 2σ derivatives from the regularity of u on $\operatorname{supp}\psi$. Repeating this argument with nested cutoffs inside $\mathbb{R}^3 \setminus \operatorname{supp}\chi_2$ gives

$$\chi_1\mathcal{R}(k)\chi_2 : H^{-N}(\mathbb{R}^3) \longrightarrow H^M(\mathbb{R}^3)$$

boundedly for every $M, N > 0$, starting with the bound (2.6). The Schwartz kernel theorem and Sobolev embedding give joint smoothness of $\mathcal{G}_k(x, y)$ off the diagonal as claimed. □

The real white noise is an isonormal Gaussian process on real $L^2(B)$, extended complex linearly for complex integrands. Thus

$$\mathbb{E}[W(h)W(j)] = \int_B hj, \qquad \mathbb{E}[W(h)\overline{W(j)}] = \int_B h\overline{j}.$$

Throughout this section, let $\mu \in L^\infty(\mathbb{R}^3)$ be nonnegative with $\operatorname{supp}\mu \subset K$. As a consequence of Lemma 2.2, we establish the well-posedness of the direct scattering problem with white noise source.

Theorem 2.3. *Let $0 < \sigma < 1$ and $k \notin \Sigma$. For the given real white noise, there exists a unique outgoing random distribution solving (1.1)–(1.2) in the class specified above. It is the centered Gaussian random distribution $u = \mathcal{R}(k)(\sqrt{\mu}\dot{W})$ and, on one event of probability one,*

$$u \in H^t_{\mathrm{loc}}(\mathbb{R}^3) \qquad \text{for every } t < 2\sigma - \frac{3}{2}.$$

The solution has a smooth modification on $\mathbb{R}^3 \setminus K$ given by

$$u(x, k) = \int_K \mathcal{G}_k(x, y)\sqrt{\mu}(y)\,dW(y), \qquad x \notin K. \tag{2.8}$$

Proof. Define the source on test functions by $\langle \sqrt{\mu}\dot{W}, \phi\rangle = \int_B \sqrt{\mu}\phi\, dW$. For an orthonormal basis $\{e_j\}$ of real $L^2(B)$, the white-noise isometry and Parseval's identity give, for every $b > 3/2$,

$$\sum_j \|\sqrt{\mu}e_j\|^2_{H^{-b}(\mathbb{R}^3)} = \frac{\|\sqrt{\mu}\|^2_{L^2(B)}}{(2\pi)^3}\int_{\mathbb{R}^3}(1+|\xi|^2)^{-b}\,d\xi < \infty.$$

Thus the Gaussian series with coefficients $\int_B e_j\, dW$ converges in $L^2(\Omega; H^{-b}(\mathbb{R}^3))$ and almost surely in H^{-b}, and defines $\sqrt{\mu}\dot{W}$ as a distribution supported in K. The realizations for a countable sequence $b \downarrow 3/2$ agree as distributions. Lemma 2.2 then constructs u, showing that it is centered Gaussian and gives the stated local regularity for all $t < 2\sigma - 3/2$ using a countable exhaustion of $\mathbb{R}^3$. It also gives for each compactly supported smooth cutoff χ

$$\mathbb{E}\,\|\chi u\|^2_{H^{2\sigma-b}} \le C_{b,\chi}\,\|\sqrt{\mu}\|^2_{L^2(B)}\,.$$

On every compact set disjoint from K, all derivatives of the kernel in (2.8) are uniformly square integrable in y after multiplication by $\sqrt{\mu}$. The isometry therefore gives finite expected Sobolev norms of every integer order on compact exterior subsets. Sobolev embedding and a countable exhaustion yield a common smooth modification. Stochastic Fubini, first against a countable dense family of test functions, identifies this field with the distribution $\mathcal{R}(k)(\sqrt{\mu}\dot{W})$.

Pathwise, the effective source $\sqrt{\mu}\dot{W} - Vu$ is compactly supported and has finite order. The free-kernel estimates in the proof of Lemma 2.2 give $u = O(|x|^{-1})$ and (1.2) on the same probability-one event. The deterministic uniqueness in Lemma 2.2 applied for each realization of the prescribed white noise proves the uniqueness. □

2.2. **The continuous mild solutions.** The stochastic representation in Theorem 2.3 is defined outside K for every $0 < \sigma < 1$. We now construct a continuous realization on $\mathbb{R}^3$ when $\sigma > 3/4$, following the constructive approach of [1, Theorem 2.7] and its fractional adaptation in [18, Section 2.2]. For convenience we denote

$$W(h) := \int_B h\, dW.$$

We first give a precise approximation of the random source. Recall that a bounded operator $T : H_1 \to H_2$ between Hilbert spaces is Hilbert–Schmidt if

$$\|T\|^2_{\mathrm{HS}} := \sum_{j=1}^{\infty}\|Te_j\|^2_{H_2} < \infty$$

for an orthonormal basis $\{e_j\}$ of H_1. The sum does not depend on the basis. Choose a real-valued orthonormal basis $\{e_j\}$ of $L^2(B)$. In the present setting, the map $J : L^2(B) \to H^{-b}(\mathbb{R}^3)$, $Jh = \sqrt{\mu}h$, satisfies for $b > 3/2$

$$\|J\|^2_{\mathrm{HS}} = \frac{\|\sqrt{\mu}\|^2_{L^2(B)}}{(2\pi)^3}\int_{\mathbb{R}^3}(1+|\xi|^2)^{-b}\,d\xi < \infty. \tag{2.9}$$

Indeed, using Fourier transformation, Tonelli's theorem and Parseval's identity in $L^2(B)$ we have

$$\sum_j \|\sqrt{\mu}e_j\|^2_{H^{-b}} = \frac{1}{(2\pi)^3}\int_{\mathbb{R}^3}(1+|\xi|^2)^{-b}\sum_j\left|\int_B \sqrt{\mu}(y)e_j(y)e^{-iy\cdot\xi}\,dy\right|^2 d\xi,$$

which equals the right-hand side of (2.9). Hence, the partial sums $\sum_{j=1}^n W(e_j)Je_j$ converge in $L^2(\Omega; H^{-b}(\mathbb{R}^3))$ as $n \to \infty$. The pairing of the random source $\sqrt{\mu}\dot{W}$ with a test function ϕ is $W(\sqrt{\mu}\phi)$, which defines $\sqrt{\mu}\dot{W}$ as a random distribution.

Proposition 2.4. *Let $\{B_{n,j}\}_{j=1}^{N_n}$ be finite measurable partitions of B into sets of positive measure such that $\max_j \operatorname{diam}(B_{n,j}) \to 0$. Define*

$$\dot{W}_n = \sum_{j=1}^{N_n} \frac{W(\mathbf{1}_{B_{n,j}})}{|B_{n,j}|} \mathbf{1}_{B_{n,j}}, \qquad u_n = \mathcal{R}(k)(\sqrt{\mu}\dot{W}_n),$$

where $\mathbf{1}_{B_{n,j}}(x)$ is the characteristic function, and $W(\mathbf{1}_{B_{n,j}}) = \int_{B_{n,j}} \mathrm{d}W$. For every $b > 3/2$ and $\eta \in C_c^\infty(\mathbb{R}^3)$ we have

$$\mathbb{E}\,\|\eta(u_n - u)\|^2_{H^{2\sigma-b}(\mathbb{R}^3)} \longrightarrow 0. \tag{2.10}$$

Moreover, for every compact $Q \Subset \mathbb{R}^3 \setminus K$ and integer $\ell \geq 0$ we have

$$\mathbb{E}\,\|u_n - u\|^2_{C^\ell(Q)} \longrightarrow 0. \tag{2.11}$$

Proof. Since

$$\mathbb{E}\left\|\sqrt{\mu}\dot{W}_n\right\|^2_{L^2(B)} = \sum_{j=1}^{N_n} \frac{1}{|B_{n,j}|} \int_{B_{n,j}} \mu(y)\,dy < \infty,$$

each regularized random source is an L^2 function almost surely. Let P_n be the orthogonal projection in $L^2(B)$ onto the cellwise constant functions. Uniform continuity first gives $P_n h \to h$ for every continuous function h on $\overline{B}$. Then by density and $\|P_n\| \leq 1$ we obtain the strong convergence on $L^2(B)$. The white-noise isometry implies

$$\mathbb{E}\left\|\sqrt{\mu}\dot{W}_n - \sqrt{\mu}\dot{W}\right\|^2_{H^{-b}} = \|J(P_n - I)\|^2_{\mathrm{HS}} = \|(P_n - I)J^*\|^2_{\mathrm{HS}} \longrightarrow 0.$$

To see the last limit, we take the sum over an orthonormal basis $\{f_j\}$ of $H^{-b}(\mathbb{R}^3)$. For each j, the term $\|(P_n - I)J^* f_j\|^2_{L^2(B)}$ tends to zero and is bounded by $\|J^* f_j\|^2_{L^2(B)}$, whose sum is finite by (2.9). The localized resolvent estimates in Lemma 2.2 prove (2.10). For the exterior estimate, choose $\zeta \in C_c^\infty(\mathbb{R}^3)$ equal to one near K and supported away from Q. Off-diagonal smoothness of the Green function gives uniform H^b bounds for $\zeta(y)\partial_x^\alpha \mathcal{G}_k(x,y)$ with $x \in Q$, $|\alpha| \leq \ell$. Duality therefore bounds the exterior C^ℓ norm by the H^{-b} norm of the source difference, which proves (2.11). □

Proposition 2.5. *Let $3/4 < \sigma < 1$ and $k \notin \Sigma$. The solution in Theorem 2.3 has a continuous modification satisfying, on one event of probability one,*

$$u(x) = \int_K G_{\sigma,k}(x-y)\sqrt{\mu(y)}\,dW(y) - \int_B G_{\sigma,k}(x-y)V(y)u(y)\,dy, \qquad x \in \mathbb{R}^3. \tag{2.12}$$

For the same white noise, this field is unique up to indistinguishability among continuous random fields satisfying (2.12). The regularized solutions u_n in Proposition 2.4 and u have common continuous versions such that, for every bounded cube Q, $1 \leq p < \infty$, and $0 < \gamma < 2\sigma - 3/2$,

$$\mathbb{E}\,\|u_n - u\|^p_{C^{0,\gamma}(Q)} \longrightarrow 0. \tag{2.13}$$

Moreover,

$$\sup_n \mathbb{E}\,\|u_n\|^p_{C^{0,\gamma}(Q)} + \mathbb{E}\,\|u\|^p_{C^{0,\gamma}(Q)} \leq C\,\|\sqrt{\mu}\|^p_{L^\infty}. \tag{2.14}$$

The constant depends only on the fixed $\sigma, k, r, q, K, B, Q, p$, and γ.

Proof. Let P_n be the orthogonal projections from the proof of Proposition 2.4, extended complex linearly. They satisfy $\|P_n\| \leq 1$ and $P_n h \to h$ in $L^2(B)$; the partitions need not be nested. Set

$$h_x(y) = \sqrt{\mu(y)}G_{\sigma,k}(x-y), \qquad w_n(x) = W(P_n h_x), \qquad w(x) = W(h_x).$$

A direct computation on the partition cells gives $w_n = \mathcal{R}_0(k)(\sqrt{\mu}\dot{W}_n)$. In this identity P_n acts on the whole product $\sqrt{\mu(y)}G_{\sigma,k}(x-y)$. The kernel estimates in [18, Lemma 2.1] give, on every bounded cube Q and for $0 < \beta < 2\sigma - 3/2$,

$$\sup_{x\in Q} \|G_{\sigma,k}(x-\cdot)\|_{L^2(B)} \le C, \qquad \|G_{\sigma,k}(x-\cdot) - G_{\sigma,k}(z-\cdot)\|_{L^2(B)} \le C_\beta |x-z|^\beta. \tag{2.15}$$

Thus $\sup_{x\in Q}\|h_x\|_2 \le C\left\|\sqrt{\mu}\right\|_\infty$ and $\|h_x - h_z\|_2 \le C_\beta \left\|\sqrt{\mu}\right\|_\infty |x-z|^\beta$. The complex linear extension of the real white noise satisfies $\mathbb{E}|W(h)|^2 = \|h\|_2^2$ and, for each finite $s \ge 2$, $\mathbb{E}|W(h)|^s \le C_s \|h\|_2^s$.

Fix p and γ as in the statement. Choose $\gamma < \beta < 2\sigma - 3/2$, take a finite $s \ge \max\{2, p\}$ sufficiently large, and then choose t such that

$$\gamma + 3/s < t < \beta < 1.$$

Each w_n is a finite Gaussian linear combination of deterministic β-Hölder functions on Q, and is therefore a strongly measurable $W^{t,s}(Q)$-valued random variable. For every $x, z \in Q$, strong convergence of P_n gives convergence to zero of the s-th moments of both $(w_n - w_j)(x)$ and $(w_n - w_j)(x) - (w_n - w_j)(z)$ as $n, j \to \infty$. Moreover,

$$\begin{aligned}\mathbb{E}\,|(w_n - w_j)(x) - (w_n - w_j)(z)|^s &\le C_s \|(P_n - P_j)(h_x - h_z)\|_2^s \\ &\le C\left\|\sqrt{\mu}\right\|_\infty^s |x-z|^{s\beta}.\end{aligned}$$

The corresponding integrand in the fractional Sobolev seminorm is bounded by $C\left\|\sqrt{\mu}\right\|_\infty^s |x - z|^{s(\beta-t)-3}$, which is integrable on $Q \times Q$. The pointwise moments are uniformly bounded by $C\left\|\sqrt{\mu}\right\|_\infty^s$. Tonelli's theorem and dominated convergence therefore imply

$$\mathbb{E}\,\|w_n - w_j\|_{W^{t,s}(Q)}^s \longrightarrow 0, \qquad \sup_n \mathbb{E}\,\|w_n\|_{W^{t,s}(Q)}^s \le C\left\|\sqrt{\mu}\right\|_\infty^s.$$

Since $W^{t,s}(Q)$ is separable, the sequence converges in $L^s(\Omega; W^{t,s}(Q))$. The continuous embedding $W^{t,s}(Q) \hookrightarrow C^{0,\gamma}(Q)$, [8, Theorem 8.2], gives a continuous representative of its limit. At each fixed x, evaluation of this limit agrees with $W(h_x)$, because $w_n(x) \to W(h_x)$ in $L^2(\Omega)$ by the white-noise isometry. We use w for this continuous modification. Consequently,

$$\mathbb{E}\,\|w_n - w\|_{C^{0,\gamma}(Q)}^p \longrightarrow 0, \qquad \sup_n \mathbb{E}\,\|w_n\|_{C^{0,\gamma}(Q)}^p + \mathbb{E}\,\|w\|_{C^{0,\gamma}(Q)}^p \le C\left\|\sqrt{\mu}\right\|_\infty^p,$$

where the p-th moment statements follow from $p \le s$. The limits obtained on different cubes and for different admissible parameters agree at a countable dense set and hence everywhere by continuity. A countable exhaustion by cubes and a sequence of exponents increasing to $2\sigma - 3/2$ select common versions. These random variables are strongly measurable in the Hölder norm: they are continuous images of strongly measurable variables in the separable space $W^{t,s}(Q)$.

It remains to transfer these properties to the perturbed equation. Let

$$\mathcal{K}_k v = (\mathcal{R}_0(k)(Vv))|_B.$$

Its kernel $G_{\sigma,k}(x-y)V(y)$ is square integrable on $B \times B$ by (2.15), so $\mathcal{K}_k$ is compact on $L^2(B)$. Lemma 2.2 gives a bounded inverse of $I + \mathcal{K}_k$: an element of its kernel produces the homogeneous outgoing solution $-\mathcal{R}_0(k)(Vv)$, whose restriction to B is v, and hence must vanish. Set

$$\begin{aligned} v_n &= (I + \mathcal{K}_k)^{-1}(w_n|_B), \quad v = (I + \mathcal{K}_k)^{-1}(w|_B), \\ u_n &= w_n - \mathcal{R}_0(k)(Vv_n), \quad \widetilde{u} = w - \mathcal{R}_0(k)(Vv). \end{aligned}$$

The expression for u_n agrees with the regularized solution by the Lippmann–Schwinger equation, and $\widetilde{u}|_B = v$ in $L^2(B)$ almost surely. Cauchy–Schwarz and (2.15) give

$$\|\mathcal{R}_0(k)(Vv)\|_{C^{0,\gamma}(Q)} \le C\|v\|_{L^2(B)}, \qquad \|v\|_{L^2(B)} \le C\|w\|_{L^2(B)}.$$

In particular, $\widetilde{u}$ is continuous and satisfies (2.12). Choose a larger bounded cube Q' containing $Q \cup \overline{B}$. The same estimates, applied to differences, yield pathwise

$$\begin{aligned}\|u_n - \widetilde{u}\|_{C^{0,\gamma}(Q)} &\le \|w_n - w\|_{C^{0,\gamma}(Q)} + C\,\|w_n - w\|_{L^2(B)} \\ &\le C\,\|w_n - w\|_{C^{0,\gamma}(Q')}\,.\end{aligned}$$

The analogous bounds for u_n and $\widetilde{u}$ prove (2.13)–(2.14) with $\widetilde{u}$ in place of u. These deterministic bounded linear maps also preserve strong measurability.

Finally, take $b = 2\sigma > 3/2$ in Proposition 2.4. It gives $u_n \to u$ in $L^2(\Omega; L^2(Q))$ for every bounded cube, while the preceding argument gives convergence to $\widetilde{u}$ in the same space. Thus $\widetilde{u} = u$ as random distributions, and Theorem 2.3 supplies the equation and the outgoing condition on one event of probability one. If two continuous fields satisfy (2.12) for the same white noise, their difference d satisfies $(I + \mathcal{K}_k)(d|_B) = 0$. Thus $d = 0$ in B, and (2.12) then gives $d = 0$ everywhere. A countable dense set and continuity give equality on one event of probability one. □

3. Inverse random source problem

In this section, we establish the stability estimate for the inverse random source problem. For an open set W, let $H^\sigma(W)$ be the restriction of $H^\sigma(\mathbb{R}^3)$ to W, with the quotient norm, and let $H^{-\sigma}(W)$ be the dual of $\widetilde{H}^\sigma(W)$ which is the closure of $C_c^\infty(W)$ in $H^\sigma(\mathbb{R}^3)$.

The following local continuation estimate is the main ingredient for deriving the stability. Its proof uses the Cauchy interpolation estimate and the interior three-ball inequality for the Caffarelli–Silvestre extension in [21, Propositions 5.13 and 5.4].

Lemma 3.1. *Let $B \subset \mathbb{R}^3$ be bounded and let $W \subset \mathbb{R}^3$ be a nonempty open ball. There exists $\kappa > 0$, depending only on σ, B, W, with the following property. Suppose that $z \in H^\sigma(\mathbb{R}^3)$ satisfies*

$$\|z\|_{H^\sigma(\mathbb{R}^3)} \le M, \qquad \|z\|_{H^\sigma(W)} + \|(-\Delta)^\sigma z\|_{H^{-\sigma}(W)} \le \varepsilon, \tag{3.1}$$

where $M > 0$ and $\varepsilon \ge 0$. For every fixed $0 < \theta_0 < \sigma/\kappa$,

$$\|z\|_{L^2(B)} \le CM \left(\log\left(3 + \frac{M}{\varepsilon}\right)\right)^{-\theta_0}. \tag{3.2}$$

The right-hand side is interpreted as zero when $\varepsilon = 0$. The constant depends only on σ, B, W and the fixed θ_0.

Proof. The estimates below apply to complex-valued functions by treating their real and imaginary parts separately. Let Z be the Caffarelli–Silvestre extension [2] of z given by

$$Z(\cdot, t) = P_{\sigma,t} * z, \quad t > 0,$$

where

$$P_{\sigma,t}(x) = c_\sigma \frac{t^{2\sigma}}{(|x|^2 + t^2)^{(3+2\sigma)/2}}.$$

Here the constant c_σ is chosen such that $\int_{\mathbb{R}^3} P_{\sigma,t}\,dx = 1$. By the extension identities in [21, Section 4] we have

$$\operatorname{div}(t^{1-2\sigma}\nabla Z) = 0 \quad \text{in } \mathbb{R}^4_+$$

and

$$Z(\cdot, 0) = z, \qquad \lim_{t\to 0} t^{1-2\sigma}\partial_t Z = -a_\sigma(-\Delta)^\sigma z,$$

where a_σ is a positive constant. The trace and conormal limits hold in $L^2(\mathbb{R}^3)$ and $H^{-\sigma}(\mathbb{R}^3)$, respectively. The unit-mass kernel gives $\sup_{t>0}\|Z(\cdot,t)\|_{L^2} \le \|z\|_{L^2}$, while the extension energy identity gives

$$\int_{\mathbb{R}^4_+} t^{1-2\sigma}|\nabla Z|^2 = a_\sigma \left\|(-\Delta)^{\sigma/2} z\right\|_{L^2}^2.$$

Consequently, for each fixed $H > 0$,

$$\left\|t^{(1-2\sigma)/2}\nabla Z\right\|_{L^2(\mathbb{R}^4_+)} + \left\|t^{(1-2\sigma)/2} Z\right\|_{L^2(\mathbb{R}^3\times(0,H))} \le C_H M. \tag{3.3}$$

Assume first that $0 < \rho := \varepsilon/M < 1$. Choose a fixed half-ball whose flat base is contained in W. The local Cauchy interpolation estimate of [21, Proposition 5.13], after translation and a fixed rescaling, gives an interior ball D_0 and $a \in (0,1)$ such that

$$\left\|t^{(1-2\sigma)/2} Z\right\|_{L^2(D_0)} \le CM\rho^a. \tag{3.4}$$

Here a is an interpolation exponent local to this proof. Indeed, both boundary terms in that proposition are bounded by $C\varepsilon$, and its bulk term is bounded by (3.3). In addition, its assumptions concern only the local extension equation, so no support condition on z is needed.

Next, for $0 < h < h_0 < 1$, we propagate (3.4) to balls above $B \times [h, 2h]$. First, we move at a fixed positive height and then descend with geometrically decreasing heights. The balls may be centered at height $5r$ with radius r, as in [21, Proposition 5.4]. During descent, take $r_{j+1} = 4r_j/5$ with a fixed horizontal center. The center distance is r_j, so the next ball lies in the preceding ball of radius $2r_j$, while each ball of radius $4r_j$ stays above $t = 0$. The balls remain in a fixed slab of the upper half-space, and the number of steps satisfies

$$N(h) \le N_0 + N_1 \log(1/h).$$

The constants are uniform over the bounded target set. The scaled three-ball inequality and (3.3) yield, for every final ball D_h of radius comparable to h,

$$\left\|t^{(1-2\sigma)/2} Z\right\|_{L^2(D_h)} \le CM\rho^{a\alpha^{N(h)}} \le CM\rho^{ch^\kappa},$$

where $\alpha \in (0,1)$ and $c, \kappa > 0$ are fixed. Here the multiplicative constants remain bounded because the normalized iteration involves the sum $1 + \alpha + \alpha^2 + \cdots$. A covering by at most Ch^{-3} such balls therefore gives

$$\left\|t^{(1-2\sigma)/2} Z\right\|_{L^2(B\times(h,2h))} \le CMh^{-3/2}\rho^{ch^\kappa}. \tag{3.5}$$

To recover the trace, for almost every $t_0 \in [h, 2h]$ the weighted Cauchy–Schwarz inequality and (3.3) imply

$$\|z - Z(\cdot, t_0)\|_{L^2(B)} \le \left(\int_0^{t_0} t^{2\sigma-1}\, dt\right)^{1/2} \left\|t^{(1-2\sigma)/2}\partial_t Z\right\|_{L^2(B\times(0,t_0))} \le CMh^\sigma. \tag{3.6}$$

This follows first between positive heights and then by the L^2 trace limit. By averaging we also obtain such a t_0 with

$$\|Z(\cdot, t_0)\|_{L^2(B)} \le Ch^{\sigma-1} \left\|t^{(1-2\sigma)/2} Z\right\|_{L^2(B\times(h,2h))},$$

which combined with (3.5) yields

$$\|z\|_{L^2(B)} \le CM\left(h^{\sigma-5/2}\rho^{ch^\kappa} + h^\sigma\right). \tag{3.7}$$

For large $L = \log(1/\rho)$, choose

$$h = \left(\frac{3\log L}{c\kappa L}\right)^{1/\kappa},$$

which is less than h_0 for all sufficiently large L. The ratio of the first term in (3.7) to the second one is

$$h^{-5/2}\exp(-cLh^\kappa) = \left(\frac{c\kappa}{3}\right)^{5/(2\kappa)} (\log L)^{-5/(2\kappa)} L^{-1/(2\kappa)}.$$

This ratio is bounded, and hence

$$\|z\|_{L^2(B)} \le CM \left(\frac{\log L}{L}\right)^{\sigma/\kappa}.$$

With $L = \log(3 + M/\varepsilon)$, the preceding estimate gives

$$\|z\|_{L^2(B)} \le CM \left(\frac{\log(\mathrm{e} + L)}{L}\right)^{\sigma/\kappa}, \tag{3.8}$$

for small positive ε/M, with a constant depending only on σ, B, W. When ε/M is bounded below, the same estimate follows from $\|z\|_{L^2(B)} \le M$ after increasing C. For each fixed $0 < \theta_0 < \sigma/\kappa$, combining (3.8) and the elementary bound

$$\left(\frac{\log(\mathrm{e} + L)}{L}\right)^{\sigma/\kappa} \le CL^{-\theta_0}, \qquad L \ge \log 3,$$

we prove (3.2). The constant in this last bound may increase as θ_0 approaches σ/κ. If $\varepsilon = 0$, apply the positive-error estimates with upper bounds $\varepsilon_j \downarrow 0$ and pass to the limit. The proof is complete. □

Denoting

$$b_{\sigma,k}(\xi) = \frac{|\xi|^2 - k^2}{|\xi|^{2\sigma} - k^{2\sigma}} \quad (|\xi| \ne k), \qquad b_{\sigma,k}(\xi) = \frac{k^{2-2\sigma}}{\sigma} \quad \text{if } |\xi| = k,$$

which is the positive multiplier introduced in Lemma A.1. The following lemma is useful in the subsequent analysis.

Lemma 3.2. *Let $J \in H^{2-2\sigma}(\mathbb{R}^3)$ with compact support in B, and set $Z = b_{\sigma,k}(D)J$. Then*

$$\|J\|_{L^2(\mathbb{R}^3)} \le k^{2\sigma-2} \|Z\|_{L^2(B)}. \tag{3.9}$$

Moreover,

$$\|J\|_{H^{1-\sigma}(\mathbb{R}^3)} \le C \|Z\|_{L^2(B)}, \tag{3.10}$$

where C depends only on σ and k.

Proof. For $t = |\xi|^2/k^2$, the inequalities $t^\sigma \ge t$ on $[0,1]$ and $t^\sigma \le t$ on $[1,\infty)$ give

$$b_{\sigma,k}(\xi) \ge k^{2-2\sigma}. \tag{3.11}$$

The multiplier is continuous and positive, including at $|\xi| = k$, and is asymptotic to $|\xi|^{2-2\sigma}$ as $|\xi| \to \infty$. Consequently,

$$c\langle\xi\rangle^{2-2\sigma} \le b_{\sigma,k}(\xi) \le C\langle\xi\rangle^{2-2\sigma}. \tag{3.12}$$

In particular, the assumed regularity of J gives $Z \in L^2(\mathbb{R}^3)$. Parseval's identity and the support of J imply

$$k^{2-2\sigma} \|J\|_{L^2}^2 \le (2\pi)^{-3} \int_{\mathbb{R}^3} b_{\sigma,k} |\widehat{J}|^2 \, d\xi = \int_B \overline{J} Z \, dx \le \|J\|_{L^2} \|Z\|_{L^2(B)},$$

which proves (3.9). Keeping the full lower bound in (3.12) gives

$$c \|J\|_{H^{1-\sigma}}^2 \le (2\pi)^{-3} \int_{\mathbb{R}^3} b_{\sigma,k} |\widehat{J}|^2 \, d\xi = \int_B \overline{J} Z \, dx \le \|J\|_{H^{1-\sigma}} \|Z\|_{L^2(B)}.$$

The middle quantity is real and nonnegative. Dividing when $J \ne 0$ proves (3.10) and the case $J = 0$ is immediate. The proof is complete. □

For a source f supported in K, we let

$$w = \mathcal{R}(k)f \quad \text{and} \quad z = (-\Delta - k^2)w.$$

Recalling $V = q + k^{2\sigma}(1-r)$, denote the effective source by

$$H = f - Vw.$$

Then we have $w = \mathcal{R}_0(k)H$, $z = b_{\sigma,k}(D)H$, and the following identity

$$((-\Delta)^\sigma - k^{2\sigma})z = (-\Delta - k^2)H \tag{3.13}$$

in the sense of distributions.

Lemma 3.3. *Let $0 < \sigma < 1$, $m > 3/2$, and let W_0, W_1 be nonempty balls with $W_0 \Subset W_1 \Subset \mathbb{R}^3 \setminus \overline{B}$. Suppose that $H \in H^m(\mathbb{R}^3)$ is supported in K, that $z = b_{\sigma,k}(D)H$, and that*

$$\|H\|_{H^m} \le M, \qquad \|z\|_{H^\sigma(W_1)} \le \varepsilon,$$

where $M > 0$ and $\varepsilon \ge 0$. Set $a = (2 - \sigma - m)_+$ and let κ be as in Lemma 3.1 for B and $W = W_0$. For every fixed $0 < \theta_0 < \sigma/\kappa$, one has

$$\|H\|_{H^{1-\sigma}} \le CM \left(\log\left(3 + \frac{M}{\varepsilon}\right)\right)^{-\theta_0(m-1+\sigma)/(m-1+\sigma+a)}. \tag{3.14}$$

The right-hand side is zero when $\varepsilon = 0$. The constant depends on the fixed parameters, balls and θ_0.

Proof. The identity (3.13) gives $(-\Delta)^\sigma z = k^{2\sigma} z$ on W_1. If $a = 0$, then $m \ge 2 - \sigma > 2 - 2\sigma$, so $H \in H^{2-2\sigma}$ and $\|z\|_{H^\sigma(\mathbb{R}^3)} \le CM$. The local equation bounds the conormal datum in $H^{-\sigma}(W_0)$ by $C\varepsilon$. Lemma 3.1 and (3.10), applied directly to $J = H$, give the asserted estimate in this case; using (3.8) gives (3.19) below with $a = 0$.

Suppose now that $a > 0$. Then $0 < \sigma < 1/2$ and $3/2 < m < 2 - \sigma < 2$, and in particular $0 < m - 1 + \sigma < 1$. Choose an even nonnegative function $\phi \in C_c^\infty(B_1(0))$ with $\int \phi = 1$. Fix $e_0 \in (0,1)$ such that

$$K + \overline{B}_{e_0}(0) \Subset B, \qquad \overline{W}_0 + \overline{B}_{e_0}(0) \Subset W_1.$$

For $0 < e \le e_0$, set

$$\phi_e(x) = e^{-3}\phi(x/e), \qquad H_e = \phi_e * H, \qquad z_e = \phi_e * z = b_{\sigma,k}(D)H_e.$$

Thus $H_e \in C_c^\infty(B)$. Evenness and unit mass give

$$|1 - \widehat{\phi}(e\xi)| \le C \min\{e^2|\xi|^2, 1\} \le Ce^{m-1+\sigma}|\xi|^{m-1+\sigma}.$$

This inequality, the rapid decay of $\widehat{\phi}$, and (3.12) yield

$$\|H - H_e\|_{H^{1-\sigma}} \le CMe^{m-1+\sigma}, \qquad \|z_e\|_{H^\sigma(\mathbb{R}^3)} \le CMe^{-a}. \tag{3.15}$$

For the second bound, the total multiplier order is $\sigma + 2 - 2\sigma = 2 - \sigma = m + a$.

Choose $\psi \in C_c^\infty(W_1)$ equal to one near $\overline{W}_0 + \overline{B}_{e_0}(0)$. On W_0, $z_e = \phi_e * (\psi z)$, and

$$\|\psi z\|_{H^\sigma(\mathbb{R}^3)} \le C \|z\|_{H^\sigma(W_1)}.$$

Convolution is uniformly bounded on H^σ. Also,

$$((-\Delta)^\sigma - k^{2\sigma})z_e = (-\Delta - k^2)H_e = 0 \quad \text{on } W_0,$$

because H_e remains supported in B. Consequently,

$$\|z_e\|_{H^\sigma(W_0)} + \|(-\Delta)^\sigma z_e\|_{H^{-\sigma}(W_0)} \le C\varepsilon. \tag{3.16}$$

For sufficiently small positive ε/M, let $L = \log(3 + M/\varepsilon)$. Applying Lemma 3.1 with global bound CMe^{-a} and local bound $C\varepsilon$ gives

$$\|z_e\|_{L^2(B)} \le CMe^{-a}L^{-\theta_0} \tag{3.17}$$

and

$$\|z_e\|_{L^2(B)} \le CMe^{-a}\left(\frac{\log(\mathrm{e}+L)}{L}\right)^{\sigma/\kappa}. \tag{3.18}$$

Indeed, $e^{-a} \ge 1$, and the logarithm corresponding to these two bounds is at least L minus a fixed constant. The function $\log(\mathrm{e}+L)/L$ is positive and decreasing for $L > 0$; fixed changes in L can therefore be absorbed in C when L is large.

Since H_e is smooth and compactly supported, it satisfies the $H^{2-2\sigma}$ assumption of Lemma 3.2. Apply (3.10) with $J = H_e$ and $Z = z_e$. Together with (3.15) and (3.17), this gives

$$\|H\|_{H^{1-\sigma}} \le CM(e^{m-1+\sigma} + e^{-a}L^{-\theta_0}).$$

Choosing $e = L^{-\theta_0/(m-1+\sigma+a)} \le e_0$ proves (3.14). The same balance with $\Phi(L) = (\log(\mathrm{e}+L)/L)^{\sigma/\kappa}$ and $e = \Phi(L)^{1/(m-1+\sigma+a)}$ gives the useful refined bound

$$\|H\|_{H^{1-\sigma}} \le CM\Phi(L)^{(m-1+\sigma)/(m-1+\sigma+a)}. \tag{3.19}$$

For non-small ε/M, both estimates follow from the a priori bound by enlarging C. For $\varepsilon = 0$, use positive upper bounds tending to zero. The proof is complete. □

Now we complete the proof of Theorem 1.1.

Proof. Write $\eta = \mu_1 - \mu_2$ and $\Delta F = F_1 - F_2$. By Lemma A.2, choose a fixed $g \in C_c^\infty(O;\mathbb{C})$ such that

$$v = \mathcal{T}g \quad \text{and} \quad \|v-1\|_{C(K)} \le \frac{1}{4}. \tag{3.20}$$

The function v is smooth near K and satisfies $|v| \ge 3/4$ on K. Set

$$f = \eta v, \qquad w = \mathcal{R}(k)f.$$

Then $\|f\|_{H^m} \le CE$. The bilinear white-noise isometry gives

$$F(x,z;k) = \int_K \mu(y)\mathcal{G}_k(x,y)\mathcal{G}_k(z,y)\,dy, \qquad x,z \in O. \tag{3.21}$$

Reciprocity and Fubini's theorem imply

$$w(x) = \int_O \Delta F(x,\zeta;k)g(\zeta)\,d\zeta, \qquad \|w\|_{L^2(O)} \le C\delta. \tag{3.22}$$

Choose concentric balls $W_0 \Subset W_1 \Subset O_1 \Subset O_2 \Subset O$. The smoothness of the Green kernel off the diagonal gives

$$\|w\|_{H^4(O_2)} \le C\,\|f\|_{L^2(K)} \le CE. \tag{3.23}$$

After multiplication by a cutoff supported in O_2 and equal to one near $\overline{O}_1$, interpolation between L^2 and H^4 gives

$$\|w\|_{H^{\sigma+2}(O_1)} \le CE^{1-b}\delta^b, \tag{3.24}$$

where $b = (2-\sigma)/4 > 0$. Put $H = f - Vw$ and $z = (-\Delta - k^2)w = b_{\sigma,k}(D)H$. Lemma 2.2 and the smoothness of V imply $\|H\|_{H^m} \le CE$ and $\operatorname{supp} H \subset K$. Moreover,

$$\|z\|_{H^\sigma(W_1)} \le CE^{1-b}\delta^b. \tag{3.25}$$

For small positive δ/E, the logarithms $\log(3+c(E/\delta)^b)$ and $\log(3+E/\delta)$ are comparable. Lemma 3.3, with $M = CE$, therefore yields

$$\|H\|_{H^{1-\sigma}} \le CE\left(\log\left(3+\frac{E}{\delta}\right)\right)^{-\theta_0(m-1+\sigma)/(m-1+\sigma+a)}, \qquad a = (2-\sigma-m)_+.$$

The identity $w = \mathcal{R}_0(k)H$ and the localized mapping property in Lemma 2.1 give

$$f = H + V\mathcal{R}_0(k)H, \qquad \|f\|_{H^{1-\sigma}} \le C\,\|H\|_{H^{1-\sigma}}.$$

By continuity, v is nonzero in a neighborhood of K. Choose $\chi \in C_c^\infty(\mathbb{R}^3)$ equal to one near K and supported in that neighborhood. The function χ/v, extended by zero, is smooth and compactly supported, and $\eta = (\chi/v)f$. Thus

$$\|\eta\|_{L^2} \le \|\eta\|_{H^{1-\sigma}} \le CE\left(\log\left(3+\frac{E}{\delta}\right)\right)^{-\theta_0(m-1+\sigma)/(m-1+\sigma+a)}. \tag{3.26}$$

Since $0 < 1-\sigma < 3/2 < m$, Fourier decomposition at radius $\rho \ge 1$ gives

$$\|\eta\|_{L^\infty} \le C\rho^{1/2+\sigma}\|\eta\|_{H^{1-\sigma}} + C\rho^{3/2-m}\|\eta\|_{H^m}.$$

For $\eta \ne 0$, choose

$$\rho = \left(\frac{\|\eta\|_{H^m}}{\|\eta\|_{H^{1-\sigma}}}\right)^{1/(m-1+\sigma)} \ge 1$$

to balance the two terms. This gives

$$\|\eta\|_{L^\infty} \le C\|\eta\|_{H^{1-\sigma}}^{(m-3/2)/(m-1+\sigma)}\|\eta\|_{H^m}^{(1/2+\sigma)/(m-1+\sigma)}. \tag{3.27}$$

The case $\eta = 0$ is immediate. Combining these estimates with $\|\eta\|_{H^m} \le 2E$ proves (1.6) with

$$\theta = \theta_0\frac{m-3/2}{m-1+\sigma+a} = \theta_0\frac{m-3/2}{\max\{m-1+\sigma,1\}}.$$

If δ/E is bounded below, the result follows from the Sobolev bound after increasing C. If $\delta = 0$, use arbitrary positive upper bounds $\delta_j \downarrow 0$ in the preceding argument and pass to the limit. The proof is complete. □

Corollary 3.4. *Under the assumptions of Theorem 1.1, let κ be the constant in Lemma 3.1 for B and $W = W_0$, where W_0 is chosen in the proof of that theorem. For $\delta > 0$, put $L = \log(3 + E/\delta)$. Then*

$$\|\mu_1 - \mu_2\|_{L^\infty} \le CE\left(\frac{\log(\mathrm{e}+L)}{L}\right)^{\frac{\sigma}{\kappa}\frac{m-3/2}{\max\{m-1+\sigma,1\}}}. \tag{3.28}$$

The right-hand side is interpreted as zero when $\delta = 0$.

Proof. Repeat the preceding proof using (3.19). For small positive δ/E, both the logarithms $\log(3+c(E/\delta)^b)$ and $\log(3+E/\delta)$, and the values of $\log(\mathrm{e}+L)/L$ at these logarithms, are comparable. The resulting exponent on the $H^{1-\sigma}$ bound is $\frac{\sigma}{\kappa}\frac{m-1+\sigma}{m-1+\sigma+a}$. Interpolation in (3.27) gives (3.28). Non-small and zero discrepancies are handled exactly as in that proof. □

The refined estimate implies the pure logarithmic estimates of Theorem 1.1 for every fixed $0 < \theta_0 < \sigma/\kappa$.

We next show that no uniform Hölder estimate holds on the full admissible class. Specifically, there is no $\alpha > 0$ and finite constant C such that

$$\|\mu_1 - \mu_2\|_{L^\infty} \le C\|F_1 - F_2\|_{L^2(O\times O)}^\alpha, \qquad \mu_1, \mu_2 \in \mathcal{A}. \tag{3.29}$$

To verify this, assume the Hölder stability holds for some constants $\alpha > 0$ and $C > 0$. Choose nonnegative $\varphi, \psi \in C_c^\infty(\operatorname{int} K)$ such that $\varphi = 1$ on a ball and $\psi \ge \varphi$. For integers $n \ge 1$, let

$$\mu_{1,n} = c(\psi + n^{-m}\varphi\cos(ny_1)), \qquad \mu_{2,n} = c(\psi - n^{-m}\varphi\cos(ny_1)).$$

These functions are nonnegative. The Fourier translation formula shows that $\|n^{-m}\varphi\cos(ny_1)\|_{H^m}$ is bounded independently of n. Choose a fixed $c > 0$ sufficiently small such that both strengths belong to $\mathcal{A}$. Letting $\eta_n = \mu_{1,n} - \mu_{2,n}$, we have

$$\|\eta_n\|_{L^\infty} \ge c_0 n^{-m} \tag{3.30}$$

for all sufficiently large n, since $\cos(ny_1)$ attains the value one in the ball where $\varphi = 1$.

By (3.21), the correlation difference is

$$\Delta F_n(x,\zeta) = 2cn^{-m}\int_K \mathcal{G}_k(x,y)\mathcal{G}_k(\zeta,y)\varphi(y)\cos(ny_1)\,dy.$$

All source-variable derivatives of the kernel factor are uniformly bounded on $\overline{O}\times\overline{O}\times\operatorname{supp}\varphi$. Repeated integration by parts in y_1 therefore gives, for every integer $N\geq 1$,

$$\|\Delta F_n\|_{L^2(O\times O)} \leq C_N n^{-m-N}. \tag{3.31}$$

If (3.29) held, then (3.30)–(3.31) would imply $c_0 n^{-m} \leq CC_N^{\alpha} n^{-\alpha(m+N)}$. Choosing N with $\alpha(m+N) > m$ and letting $n\to\infty$ we obtain a contradiction.

We now prove Theorem 1.2 under the assumption that $\mu_1-\mu_2$ has a fixed sign.

Proof. Using Lemma A.2 we may choose $g\in C_c^\infty(O;\mathbb{C})$ such that $v=\mathcal{T}g$ satisfies $\|v-1\|_{C(K)}\leq 1/4$. Thus, on the set K we have $\Re v\geq 3/4$ and $|\Im v|\leq 1/4$, which gives $\Re(v^2)\geq 1/2$. Write $\eta=\mu_1-\mu_2$. Since the sign of η is fixed, we obtain

$$\frac{1}{2}\|\eta\|_{L^1} \leq \left|\int_K \eta(y)v(y)^2\,dy\right| = \left|\int_{O\times O}(F_1-F_2)(x,z;k)g(x)g(z)\,dx\,dz\right| \leq \|g\|_{L^2(O)}^2\,\delta.$$

Thus we prove the first estimate in (1.8). Further, combining the Fourier inversion and the estimate $|\widehat{\eta}|\leq\|\eta\|_{L^1}$, we have for $R\geq 1$

$$\|\eta\|_{L^\infty} \leq C(R^3\|\eta\|_{L^1} + R^{3/2-m}\|\eta\|_{H^m}).$$

Since $\|\eta\|_{H^m}\leq 2E$, optimizing R when δ/E is small gives the second estimate in (1.8). For larger δ/E, this estimate still holds after increasing C.

To prove the sharpness of the Hölder exponent, choose y_0 in the interior of K and a nonzero smooth function $\varphi\geq 0$ in $C_c^\infty(B_1)$. For small $h>0$, consider the strength functions μ_h and ν_h with

$$\mu_h(y) = cEh^{m-3/2}\varphi((y-y_0)/h), \qquad \nu_h = 0,$$

where $c>0$ is a constant such that $\|\mu_h\|_{H^m}\leq E$. Then $\|\mu_h\|_{L^\infty}$ is comparable to $Eh^{m-3/2}$, and $\|\mu_h\|_{L^1}$ is comparable to $Eh^{m+3/2}$. Since the Green kernel is smooth and bounded on the separated set $\overline{O}\times K$, from (3.21) we have $\|F_{\mu_h}-F_{\nu_h}\|_{L^2(O\times O)}\leq CEh^{m+3/2}$, where F_{μ_h} and F_{ν_h} denote the correlation data with $\mu=\mu_h$ and $\mu=\nu_h$, respectively. Hence, any Hölder stability estimate with an exponent $\alpha>\beta$ would imply $h^{m-3/2}\leq Ch^{\alpha(m+3/2)}$ for all small h, which is a contradiction. □

4. Conclusion

We have studied an inverse random source problem for the fractional Helmholtz equation in an inhomogeneous medium. For admissible frequencies, we established well-posedness of the radiating solution with a spatial white noise source. The solution has a smooth representation outside the source region and, when the fractional order exceeds 3/4, admits a continuous mild realization.

Using two-point correlations of the field measured on an exterior open set at a single frequency, we obtained logarithmic stability for the recovery of the source strength under an a priori Sobolev bound. We have shown that a uniform Hölder estimate is unavailable for the general admissible class. When the difference between two source strengths has a fixed sign, the stability improves to Lipschitz continuity in L^1 and Hölder continuity in L^∞ with an optimal Hölder exponent. These results show that quantitative recovery is possible from fixed-frequency correlation measurements even when the fractional wave propagates through an inhomogeneous medium. The analysis combines a resolvent formulation of the direct problem with approximation and quantitative continuation for the fractional Laplacian.

Appendix A. Useful lemmas

We first establish a continuation property for the free outgoing resolvent.

Lemma A.1. *Let F be a compactly supported distribution and $w = \mathcal{R}_0(k)F$. If w and $(-\Delta)^\sigma w$ vanish on a nonempty open set, then $w = 0$ and $F = 0$.*

Proof. Following the spectral regularization in [18, Theorem 3.2], set $z = (-\Delta - k^2)w$. Cancellation of the outgoing Fourier pole gives

$$\widehat{z}(\xi) = b_{\sigma,k}(\xi)\widehat{F}(\xi), \qquad b_{\sigma,k}(\xi) = \frac{|\xi|^2 - k^2}{|\xi|^{2\sigma} - k^{2\sigma}} \quad (|\xi| \neq k). \tag{A.1}$$

The value of $b_{\sigma,k}(\xi)$ on $|\xi| = k$ is $k^{2-2\sigma}/\sigma$. The multiplier is positive and grows at most polynomially. Since $\widehat{F}$ is a smooth function with polynomial growth, we have $z \in H^{-N}(\mathbb{R}^3)$ for some $N > 0$. On the given open set,

$$z = 0 \quad \text{and} \quad (-\Delta)^\sigma z = (-\Delta - k^2)(-\Delta)^\sigma w = 0.$$

The unique continuation theorem for the fractional Laplacian [10, Theorem 1.2] gives $z = 0$. Positivity of $b_{\sigma,k}$ in (A.1) then yields $\widehat{F} = 0$ which gives $F = 0$, and hence $w = 0$. Near $\xi = 0$, these multiplier products are products of locally integrable functions; away from zero they have their usual distributional meaning. □

For $g \in C_c^\infty(O;\mathbb{C})$, define

$$(\mathcal{T}g)(y) = \int_O \mathcal{G}_k(x,y)g(x)\,dx = (\mathcal{R}(k)g)(y). \tag{A.2}$$

Lemma A.2. *Let $k \notin \Sigma$ and let O be a nonempty bounded open subset of $\mathbb{R}^3 \setminus \overline{B}$. Then*

$$\{\mathcal{T}g|_K : g \in C_c^\infty(O;\mathbb{C})\} \quad \textit{is dense in } C(K;\mathbb{C}). \tag{A.3}$$

Proof. If the density failed, the Hahn–Banach and Riesz representation theorems would give a nonzero finite complex Borel measure ν supported in K such that

$$\int_K (\mathcal{T}g)(y)\,d\nu(y) = 0 \qquad \text{for every } g \in C_c^\infty(O;\mathbb{C}).$$

Its bounded Fourier transform implies $\nu \in H^{-b}(\mathbb{R}^3)$ for every $b > 3/2$, so $w = \mathcal{R}(k)\nu$ is well-defined. Reciprocity and the separated smooth kernel give

$$\int_K (\mathcal{T}g)\,d\nu = \int_O w(x)g(x)\,dx,$$

and hence $w = 0$ on O. Since $\big((-\Delta)^\sigma - k^{2\sigma}r(x) + q(x)\big)w = \nu$, $r = 1$, and $q = 0$ on O, we also have $(-\Delta)^\sigma w = k^{2\sigma}w = 0$ on O. Moreover, we have

$$w = \mathcal{R}_0(k)(\nu - Vw),$$

with a compactly supported distribution on the right. Then Lemma A.1 yields $w = 0$, and the equation $\big((-\Delta)^\sigma - k^{2\sigma}r(x) + q(x)\big)w = \nu$ gives $\nu = 0$, which is a contradiction. □

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Conflict of Interest Statement

The authors declare that they have no conflict of interest.

Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

References

[1] G. Bao, C. Chen, and P. Li, Inverse random source scattering problems in several dimensions, SIAM/ASA J. Uncertain. Quantif., 4 (2016), 1263–1287.

[2] L. Caffarelli and L. Silvestre, An extension problem related to the fractional Laplacian, Comm. Partial Differential Equations, 32 (2007), 1245–1260.

[3] X. Cao and H. Liu, Determining a fractional Helmholtz equation with unknown source and scattering potential, Commun. Math. Sci., 17 (2019), 1861–1876.

[4] G. Covi, M. de Hoop, and M. Salo, Geometrical optics for the fractional Helmholtz equation and applications to inverse problems, Math. Models Methods Appl. Sci., 35 (2025), 2831–2868.

[5] G. Covi, T. Ghosh, A. Rüland, and G. Uhlmann, A reduction of the fractional Calderón problem to the local Calderón problem by means of the Caffarelli–Silvestre extension, J. Math. Pures Appl., 211 (2026), 103886.

[6] G. Covi, J. Railo, T. Tyni, and P. Zimmermann, Stability estimates for the inverse fractional conductivity problem, SIAM J. Math. Anal., 56 (2024), 2456–2487.

[7] S. Das, T. Ghosh, and S. Ma, Inverse scattering for the fractional Schrödinger equation, arXiv:2509.12685, 2025.

[8] E. Di Nezza, G. Palatucci, and E. Valdinoci, Hitchhiker's guide to the fractional Sobolev spaces, Bull. Sci. Math., 136 (2012), 521–573.

[9] T. Ghosh, Y.-H. Lin, and J. Xiao, The Calderón problem for variable coefficients nonlocal elliptic operators, Comm. Partial Differential Equations, 42 (2017), 1923–1961.

[10] T. Ghosh, M. Salo, and G. Uhlmann, The Calderón problem for the fractional Schrödinger equation, Anal. PDE, 13 (2020), 455–475.

[11] N. Laskin, Fractional quantum mechanics, Phys. Rev. E, 62 (2000), 3135–3145.

[12] P. Li and Y. Liang, Stability for inverse source problems of the stochastic Helmholtz equation with a white noise, SIAM J. Appl. Math., 84 (2024), 687–709.

[13] P. Li and Z. Li, An inverse random source problem for the fractional Helmholtz equation, arXiv:2602.19559, 2026.

[14] C.-L. Lin, Y.-H. Lin, and G. Uhlmann, The Calderón problem for nonlocal parabolic operators: A new reduction from the nonlocal to the local, arXiv:2308.09654v1, 2023.

[15] Y.-H. Lin, Determining both leading coefficient and source in a nonlocal elliptic equation, J. Inverse Ill-Posed Probl., 33 (2025), 207–216.

[16] C. Martinez and M. Sanz, The Theory of Fractional Powers of Operators, North-Holland Math. Stud., 187, Elsevier, Amsterdam, 2001.

[17] P. Mickan and T. Hohage, Hölder-logarithmic stability and convergence rates for an inverse random source problem, arXiv:2602.20822, 2026.

[18] D. Qiu, X. Xu, and Y. Zhao, Stability for the inverse source problem of the stochastic fractional Helmholtz equation, arXiv:2609.27383, 2026.

[19] A. Rüland, Unique continuation for fractional Schrödinger equations with rough potentials, Comm. Partial Differential Equations, 40 (2015), 77–114.

[20] A. Rüland, On single measurement stability for the fractional Calderón problem, SIAM J. Math. Anal., 53 (2021), 5094–5113.

[21] A. Rüland and M. Salo, The fractional Calderón problem: Low regularity and stability, Nonlinear Anal., 193 (2020), 111529.

[22] P. R. Stinga and J. L. Torrea, Extension problem and Harnack's inequality for some fractional operators, Comm. Partial Differential Equations, 35 (2010), 2092–2122.

[23] V. E. Tarasov and J. J. Trujillo, Fractional power-law spatial dispersion in electrodynamics, Ann. Physics, 334 (2013), 1–23.

[24] D. Zilberberg, F. Cakoni, and M. S. Vogelius, Limiting absorption principle and radiation condition for the fractional Helmholtz equation, arXiv:2602.18387, 2026.

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